Learning a Deep Structural Subspace Across Hyperspectral Scenes With Cross-Domain VAE

Minchao Ye, *Member, IEEE*, Junbin Chen, Fengchao Xiong, *Member, IEEE*, and Yuntao Qian, *Senior Member, IEEE*

***Abstract*— Hyperspectral image (HSI) classification is a small-sample-size problem due to the expensive cost of labeling. As a novel approach to this problem, cross-scene HSI classification has become a hot research topic in recent years. In cross-scene HSI classification, the scene containing enough labeled samples (called source scene) is used to benefit the classification in another scene containing a small number of training samples (called target scene). Transfer learning is a typical solution for cross-scene classification. However, many transfer learning algorithms assume an identical feature space for source and target scenes, which violates the fact that source and target scenes often lie in different feature spaces with various dimensions due to different HSI sensors. Aiming at the different feature spaces between the two scenes, we propose an end-to-end heterogeneous deep transfer learning algorithm, namely, cross-domain variational autoencoder (CDVAE). This algorithm is mainly composed of two key parts: 1) the features of the two scenes are embedded into the shared feature subspace through the two-stream variational autoencoder (VAE) to ensure that the output feature dimensions of the two scenes are identical and 2) graph regularization is used to establish the manifold constraints between source and target scenes in the shared subspace, so as to align the feature spaces. Experiments on two different cross-scene HSI datasets have proved the superior performance of the proposed CDVAE algorithm.**

***Index Terms*— Cross-domain variational autoencoder (CDVAE), cross-scene classification, heterogeneous transfer learning, hyperspectral image (HSI).**

I. Introduction

PIXEL classification is a hot application on hyperspectral images (HSIs). In recent years, many traditional machine learning methods have been used for classification. However, these methods are heavily dependent on manual feature extraction and thus are not adaptive to the specific classification task. Therefore, the development of HSI classification is severely restricted. Compared with traditional machine learning, deep learning can achieve multiscale feature expression that is adaptive to the classification task via an end-to-end training scheme. Therefore, more and more researchers apply deep transfer learning to HSI classification [1]. However, most deep learning algorithms require a huge number of training samples to achieve desirable performance. On the contrary, HSI data usually contain dozens or hundreds of bands (spectral dimensions), and it causes high-dimensional problems. At the same time, it is labor-consuming to label a large number of samples, leading to insufficient training samples. Therefore, HSI classification is usually a high-dimensional small-sample-size problem.

Fortunately, it is found that two HSIs with similar land-cover classes can share information with each other via transfer learning. In detail, the source scene (with a large number of training samples) can be used to gain the classification accuracy of the target scene (with insufficient training samples). This problem is called cross-scene HSI classification [2]. A cross-scene HSI classification problem can be well handled by transfer learning. Zhang *et al.* [3] proposed a transfer learning algorithm based on a 3-D convolutional neural network (CNN). Abundant source scene samples are adopted for pretraining, and then, the knowledge is transferred to the target scene. Lin *et al.* [4] proposed an HSI classification algorithm based on active-learning-incorporated deep transfer learning. The algorithm finds out salient samples on both source and target domains through active learning, and then, autoencoders (AEs) are established on both domains. The correlation between both domains is searched by a canonical correlation analysis (CCA). Geng *et al.* [5] proposed a cross-scene HSI classification algorithm based on deep conditional distribution adaptation networks. By stacked autoencoder (SAE), the algorithm establishes constraints between source and target scenes to unify their feature subspace, so as to complete mapping-based deep transfer learning.

From the viewpoint of general machine learning, transfer learning algorithms can be divided into four categories.

1) Instance-based deep transfer learning is based on a presumption that the linear combinations of source domain samples can be regarded as target domain samples. Based on this method, Yao and Doretto *et al.* [6] proposed a deep transfer learning method based on

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multisource AdaBoost. Xu *et al.* [7] proposed a metric transfer learning framework (MTLF).

2) Network-based deep transfer learning reuses the network architecture of source domain on target domain. In [8] and [9], network-based deep transfer learning showed good performance.
3) Adversarial-based deep transfer learning is a series of deep transfer learning algorithms based on generative adversarial networks (GANs). Using GAN can effectively transfer the knowledge of source domain. Tzeng *et al.* [10], [11] proposed a special CNN architecture and a framework called adversarial discriminative domain adaptation (ADDA).
4) Mapping-based deep transfer learning maps the data of source and target domains to a shared feature subspace through the neural network. Also, these algorithms try to reduce the difference between the two kinds of data in the subspace. The algorithms proposed by Zhang *et al.* [12] and Long *et al.* [13] verify the effectiveness of mapping-based deep transfer learning.

Transfer learning methods applied to HSI classification can be divided into two types according to whether two scenes are captured by the same sensor.

1) *Homogeneous Transfer Learning:* When two scenes are imaged by the same sensor, they have the same feature dimension (number of bands), and thus, homogeneous transfer learning can be adopted to eliminate feature shift caused by different imaging conditions [14], [15]. Currently, most transfer learning algorithms for HSIs deliver homogenous transfer learning. Li *et al.* [16] proposed a two-stage deep domain adaptation method. These two stages use maximum mean discrepancy (MMD) and pairwise loss to learn a more discriminative deep embedding space. Zhang *et al.* [17] proposed a cross-scene HSI classification method with discriminative cooperative alignment (DCA), which combines subspace alignment (SA) and distribution alignment (DA) to reduce geometric and statistical shift, and a reconstruction constraint is imposed to ensure that more discriminative information is retained. These two methods show superior performance on homogeneous datasets.
2) *Heterogeneous Transfer Learning:* In most cases, source and target scenes are imaged by various sensors; hence, they have different feature dimensions and heterogeneous transfer learning is required [18]. Zhong *et al.* [19] proposed a multiscale spectral–spatial unified network, which uses the joint probability distribution and deep multiscale spectral–spatial network for heterogeneous transfer learning. Jiang *et al.* [20] proposed a 3-D separable ResNet (3-D SRnet), which combines separable 3-D CNN and ResNet to achieve a good performance on heterogeneous HSI datasets. In general, heterogeneous transfer learning is more complex than homogeneous one, and only a few algorithms are available for HSI classification. Hence, more heterogeneous transfer learning algorithms are desired to be developed for solving a cross-scene HSI classification problem.

For performing heterogeneous transfer learning, we proposed a cross-scene HSI classification algorithm based on dual-dictionary nonnegative matrix factorization (DDNMF) in our previous work [21]. DDNMF trains two dictionaries for source and target scenes and maps the features of source and target scenes into a shared subspace. Then, graph regularization is applied to reduce the difference of features between the two scenes. Although good transfer learning results have been achieved with DDNMF, the algorithm has two shortcomings: 1) this is a shallow learning algorithm and cannot extract abstract features of deeper layers and 2) feature extraction and classification are separated. Thus, only transferability is considered, but discriminability is not guaranteed.

The disadvantages of existing algorithms include: 1) most of them are homogeneous transfer learning algorithms, which are unable to handle HSIs captured by different sensors, and 2) some existing heterogeneous transfer learning algorithms for HSIs, such as DDNMF, are not end-to-end algorithms, so they cannot guarantee both transferability and discriminability.

In order to deliver cross-scene knowledge transfer by deep neural networks for HSIs, a novel deep heterogeneous transfer learning algorithm is proposed in this work, namely, cross-domain variational autoencoder (CDVAE). In CDVAE, cross-domain feature extraction and classification tasks are concatenated through an end-to-end network, thus ensuring both transferability and discriminability. The main advantages of this work are as follows.

1) An end-to-end CDVAE algorithm is proposed to deal with heterogeneous transfer learning problem with different feature dimensions. The transfer learning network architecture is driven by the cross-domain classification task.
2) On the basis of dimension reduction, variational autoencoder (VAE) maps the input data into lower dimensional data following Gaussian distributions. Combined with the classification network, samples belong to the same land-cover class are forced to share similar Gaussian distribution parameters, while samples belong to different classes are constrained to have different Gaussian distribution parameters. Thus, the discriminability of the extracted features is improved.
3) Graph regularization loss is taken into consideration for delivering data manifold alignment between the two scenes, so as to solve the problem of domain adaptation in heterogeneous transfer learning.
4) CDVAE is not sensitive to the dimension of the output subspace, and the model has good generalization ability.

The following parts of this article are organized as follows. Section II introduces the basis of VAE and graph regularization. In Section III, a deep transfer learning algorithm CDVAE is proposed. In Section IV, the experimental results on two cross-scene HSI datasets are shown and analyzed. Finally, Section V draws the conclusion and introduces future work.

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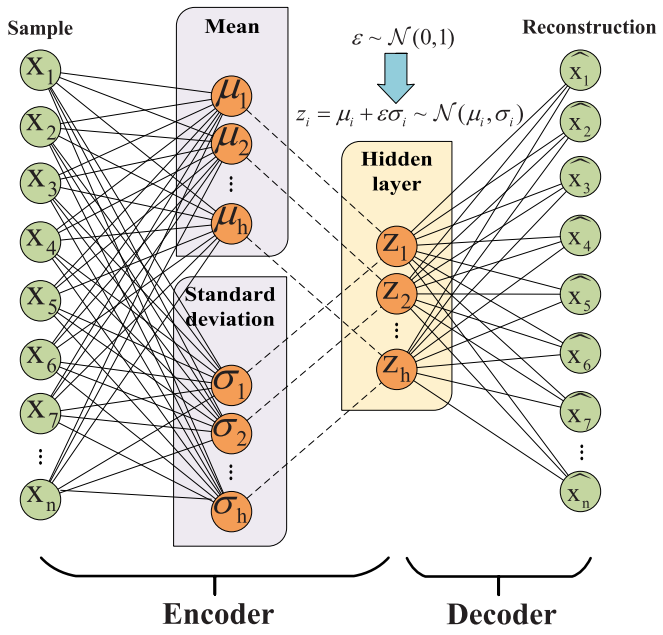


Fig. 1. Architecture of VAE.

II. Related Works

The proposed CDVAE combines VAE and graph regularization, and the details of these two basic models are briefly introduced in this section.

A. VAE

AE is an unsupervised learning algorithm for data dimension reduction. It is divided into two parts: encoder and decoder. The encoder compresses the input data into low-dimensional hidden layer data. After that, a decoder performs a reconstruction of the input data from the hidden layer data. By minimizing the difference between the input data and the decoded data, the hidden layer data maintain the information of the input data as much as possible, so as to achieve the purpose of data dimension reduction and feature extraction [22].

VAE is an algorithm proposed by Kingma and Welling [23]. Different from general AEs, VAE uses variational inference and reparameterization trick to enhance the feature extraction ability and maintain the diversity of decoded data.

Similar to general AEs, VAE is also divided into two parts, as shown in Fig. 1. The encoder encodes the input data into hidden layer data $\mathbf{z} = \text{encoder}(\mathbf{x}) \sim q(\mathbf{z}|\mathbf{x})$, through training the mean value μ and standard deviation σ, making it follow the Gaussian distribution. Then, through the decoder, the hidden layer data $\mathbf{z}$ are reconstructed to the input data $\widehat{\mathbf{x}} = \text{decoder}(\mathbf{z}) \sim p(\mathbf{x}|\mathbf{z})$. It is worth mentioning that the solid lines in Fig. 1 represent fully connected (FC) layer, while the dashed ones represent the Gaussian sampling with $z_i \sim \mathcal{N}(\mu_i, \sigma_i)$.

The loss function of VAE is composed of two parts: reconstruction loss $\mathcal{L}_{\text{rec}}$ and Kullback–Leibler (KL) divergence $\mathcal{L}_{\text{kl}}$

$$\mathcal{L}_{\text{vae}} = \mathcal{L}_{\text{rec}} + \mathcal{L}_{\text{kl}}. \tag{1}$$

Mean-square error (MSE) is utilized to measure the reconstruction loss of VAE

$$\mathcal{L}_{\text{rec}} = -\mathbb{E}_{q(\mathbf{z}|\mathbf{x})}[\log p(\mathbf{x}|\mathbf{z})] = \frac{1}{m}\sum_{i=1}^{m}(x_i - \widehat{x_i})^2. \tag{2}$$

KL divergence is used to represent the loss of difference between $q(\mathbf{z}|\mathbf{x})$ and the standard Gaussian distribution $p(\mathbf{z})$

$$\begin{aligned}\mathcal{L}_{\text{kl}} &= \text{KL}(q(\mathbf{z}|\mathbf{x})||p(\mathbf{z})) \\ &= \frac{1}{2m}\sum_{i=1}^{m}(-\log {\sigma_i}^2 + {\mu_i}^2 + {\sigma_i}^2 - 1)\end{aligned} \tag{3}$$

where KL$(\cdot)$ is the KL divergence. The derivation can be found in [23]. Since the operation of sampling from $\mathcal{N}(\mu, \sigma)$ is nonderivable, the reparameterization trick is applied. Assuming that $z = \mu + \varepsilon\sigma$, sampling $z \sim \mathcal{N}(\mu, \sigma)$ can be equivalent to sampling $\varepsilon \sim \mathcal{N}(0, 1)$. Finally, sampling results are regarded as the hidden (encoder) vector for training [24].

B. Graph Modeling

The general dimension reduction algorithm can align the dimension of input data from two scenes. However, further alignment of data manifold is still needed. As shown in Fig. 2, although two samples belong to the same class, they may have totally different input features due to different sensors. However, scene-invariant feature extraction is desired, i.e., the extracted features from two scenes are hoped to be close to each other in the output subspace. For this purpose, graph modeling is introduced, which has been adopted in many machine learning applications. Belkin and Niyogi [25] and [26] proposed a Laplacian eigenmaps method for dimension reduction and data representation. This method is based on the graph Laplacian, the Laplace–Beltrami operator on a manifold, and the heat equation, which provides an effective way for nonlinear dimension reduction. In addition, graph regularization is widely used in many studies. Cai *et al.* [27] combined graph regularization with nonnegative matrix factorization (NMF) to obtain a graph regularized nonnegative matrix factorization (GNMF) algorithm. In GNMF, a graph is created to maintain the local manifold structure of the data so that two similar samples in the input matrix remain similar in the output feature matrix. Based on GNMF, Zhang *et al.* [28] established a discriminative graph for whether two samples belong to the same class and realized nonnegative discriminative manifold learning based on this discriminative graph. Chen and Song [29] proposed a dual-graph regularized transfer sparse coding method based on graph regularization, which constructs graphs by using the interdomain and intradomain similarity. This method can effectively reduce the distribution divergence of different datasets in a facial expression recognition problem.

The data manifold structure can be modeled by the graph, where a vertex represents a sample (i.e., a pixel from HSI), and an edge represents the similarity between two samples. The weight matrix $\mathbf{W}$ of the graph can be defined in many ways, among which three most commonly used forms are as follows.

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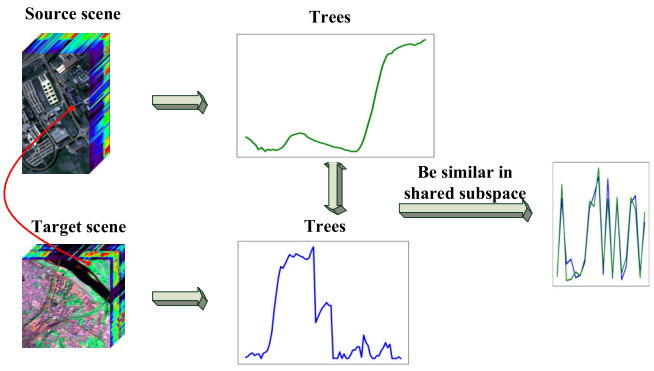


Fig. 2. Connection between source and target scenes.

1) *0–1 Weight:*

$$w_{ij} = \begin{cases} 1, & \text{class}(\mathbf{x}_i) = \text{class}(\mathbf{x}_j) \\ 0, & \text{class}(\mathbf{x}_i) \neq \text{class}(\mathbf{x}_j) \end{cases}. \tag{4}$$

2) *Heat Kernel Weight:*

$$w_{ij} = e^{-\frac{\|\mathbf{x}_i - \mathbf{x}_j\|^2}{\sigma}}. \tag{5}$$

3) *Dot-Product Weight:*

$$w_{ij} = \|\mathbf{x}_i{}^{\mathrm{T}} \mathbf{x}_j\| \tag{6}$$

where $\mathbf{x}_i$ and $\mathbf{x}_j$ are the ith and jth samples in input data, respectively.

III. Proposed CDVAE Algorithm

A. Two-Stream VAE Layer

VAE has good feature extraction capability in many cases. Dong *et al.* [30] compared VAE with traditional dimension reduction methods (such as principal component analysis (PCA) and AE) and proved that VAE is a more efficient nonlinear dimension reduction method that can represent data more stably and efficiently. In traditional feature extraction problems with sufficient training samples, VAE can solve these problems well. However, when we are faced with an insufficient sample of HSI, the single-scene VAE cannot help much. Therefore, CDVAE is proposed to perform cross-domain feature extraction to solve the problem of insufficient training samples in the target scene. As a cross-domain transfer learning algorithm, CDVAE considers the knowledge of source and target scenes at the same time and maps the samples of two different dimensions into a shared subspace, which effectively reduces the difference between two scenes. A two-stream VAE layer is the core component of CDVAE. The feature extraction in source and target scenes can be conducted by two VAE streams

$$\begin{cases} \mathbf{Z}^{\mathcal{S}} = F_{\mathcal{S}}(\mathbf{X}^{\mathcal{S}}) \\ \mathbf{Z}^{\mathcal{T}} = F_{\mathcal{T}}(\mathbf{X}^{\mathcal{T}}) \end{cases} \tag{7}$$

where $\mathbf{X}^{\mathcal{S}} \in \mathbb{R}^{p \times m}$ and $\mathbf{X}^{\mathcal{T}} \in \mathbb{R}^{q \times n}$ represent the input data of source and target scenes, respectively, m and n are their input dimensions, and p and q represent the number of source and target samples, respectively. $F_{\mathcal{S}}(\cdot)$ and $F_{\mathcal{T}}(\cdot)$ are the mapping function (the encoder of the VAE) that projects the input data to a common subspace, which are implemented by VAE. $\mathbf{Z}^{\mathcal{S}} \in \mathbb{R}^{p \times r}$ and $\mathbf{Z}^{\mathcal{T}} \in \mathbb{R}^{q \times r}$ are the feature representations of source and target scenes in the subspace, respectively, where r is their feature dimension.

For transfer learning task, two streams in (7) have to be bound so that connection between source and target scenes can be established. In this work, we impose graph regularization between source and target scenes for building the connection. According to previous studies, it is possible to use the land-cover class co-occurrence between two scenes to establish a manifold, thereby aligning the data manifold between two scenes [31], [32]. The goal is to make samples belonging to the same class keep similar to each other in the shared subspace, so a source–target discriminative graph $\mathcal{G}^{\mathcal{ST}}$ can be adopted, whose weight matrix $\mathbf{W}^{\mathcal{ST}}$ can be defined as

$$w_{i,j}^{\mathcal{ST}} = \begin{cases} \dfrac{1}{U^{\mathcal{ST}}}, & \text{class}\left(\mathbf{X}_{i,\cdot}^{\mathcal{S}}\right) = \text{class}\left(\mathbf{X}_{j,\cdot}^{\mathcal{T}}\right) \\ 0, & \text{class}\left(\mathbf{X}_{i,\cdot}^{\mathcal{S}}\right) \neq \text{class}\left(\mathbf{X}_{j,\cdot}^{\mathcal{T}}\right) \end{cases} \tag{8}$$

where $\mathbf{X}_{i,\cdot}^{\mathcal{S}}$ and $\mathbf{X}_{j,\cdot}^{\mathcal{T}}$ represent the ith and jth samples in source and target scenes, respectively, $U^{\mathrm{ST}} = \sum_{i,j} w_{i,j}^{\mathcal{ST}}$ is the normalization factor, and class$(\cdot)$ means the land-cover class of a pixel. An example can be found in Fig. 3, where $w_{i,j}^{\mathcal{ST}}$ is the weight between the ith sample in source scene and the jth sample in target scene. More details can be found on $w_{1,3}^{\mathcal{ST}}$ and $w_{5,4}^{\mathcal{ST}}$. With above graph, data manifold alignment is achieved in the shared subspace. The graph regularization loss can be defined as

$$\mathcal{L}_{\mathrm{gr}} = \sum_{i=1}^{m} \sum_{j=1}^{n} w_{i,j}^{\mathcal{ST}} \left\| \mathbf{Z}_{i,\cdot}^{\mathcal{S}} - \mathbf{Z}_{j,\cdot}^{\mathcal{T}} \right\|_2. \tag{9}$$

Combining VAE and graph regularization, a layer of CDVAE can be established. The feature embedding loss function is defined as

$$\mathcal{L}_{\mathrm{fe}} = \mathcal{L}_{\mathrm{vae}}^{\mathcal{S}} + \alpha \mathcal{L}_{\mathrm{vae}}^{\mathcal{T}} + \lambda \mathcal{L}_{\mathrm{gr}} \tag{10}$$

where $\mathcal{L}_{\mathrm{fe}}$ is the feature embedding loss function, $\mathcal{L}_{\mathrm{vae}}^{\mathcal{S}}$ is the VAE loss of source scene, $\mathcal{L}_{\mathrm{vae}}^{\mathcal{T}}$ is the VAE loss of target scene, and $\mathcal{L}_{\mathrm{gr}}$ is the graph regularization loss between source and target scenes. $\alpha = \|\mathbf{X}^{\mathcal{S}}\|_F^2 / \|\mathbf{X}^{\mathcal{T}}\|_F^2$ is a balancing parameter for the VAE losses of source and target scenes, whose setting is introduced from [21] and [33], and it works well for most cases. λ is a balancing parameter between VAE loss and graph regularization loss.

B. Architecture and Loss Function of CDVAE

CDVAE can be composed of multiple layers of the aforementioned two-stream VAE. In this work, we adopt two layers, and the final architecture is shown in Fig. 4, where the weights of the two layers are represented in blue and red, respectively. In addition, the training samples from source and target scenes are merged in the second layer, and the weights of the FC layer are shared afterward. Within the architecture of CDVAE, the shallower two VAE layers are separated into source and target streams in order to handle the heterogeneity between source and target inputs. However,

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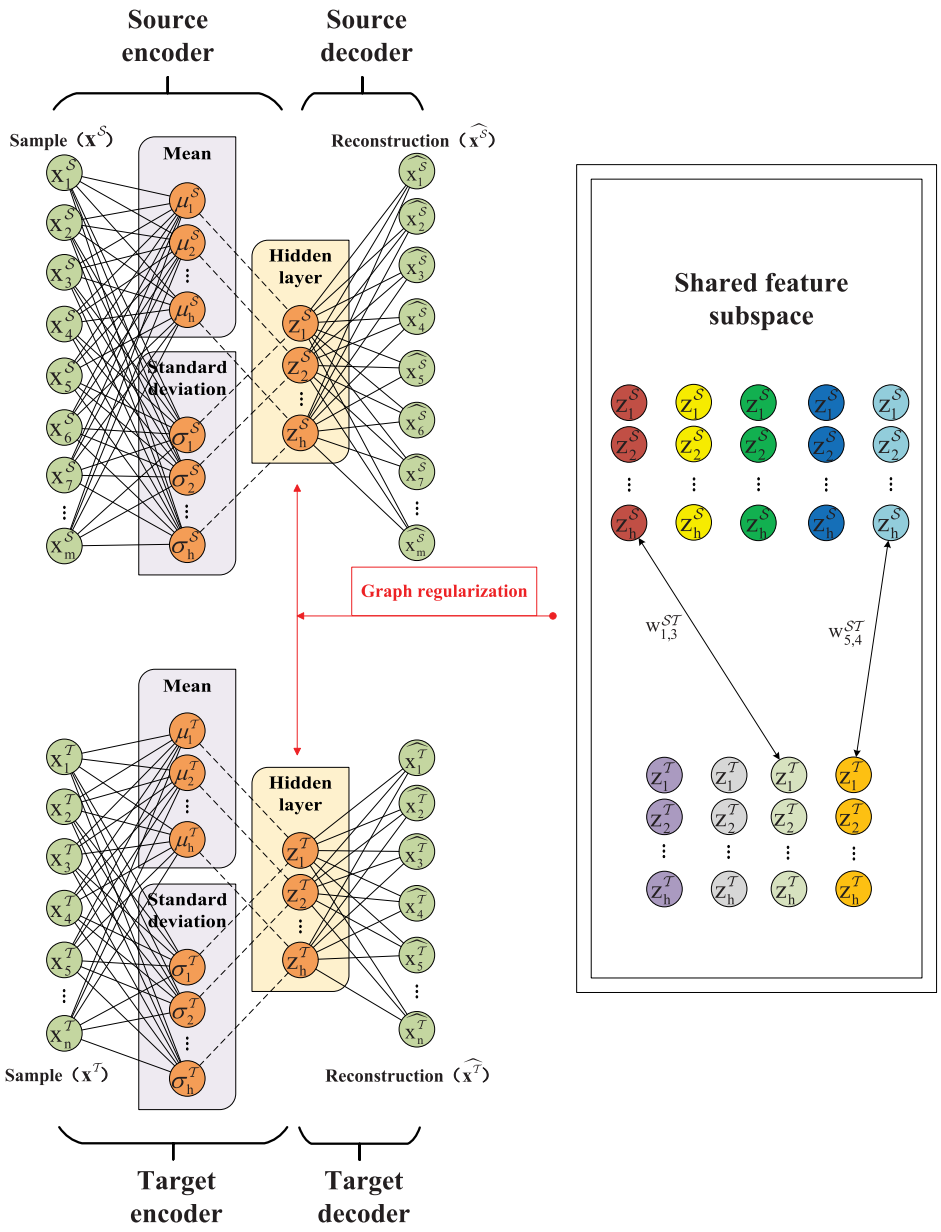


Fig. 3. Graph regularized cross-scene feature embedding based on two-stream VAE.

two streams are not exactly separated, we softly bridge them by a graph regularization technique. The deeper FC layer (i.e., output layer for classification) is shared between two scenes for guaranteeing cross-domain consistency of outputs.

The loss function of CDVAE can be written as

$$\mathcal{L}\left(\mathbf{X}^{\mathcal{S}}, \mathbf{X}^{\mathcal{T}}\right) = \sum_{i=1}^{l} \beta_i \mathcal{L}_{\mathrm{fe}_i} + \theta \mathcal{L}_c \tag{11}$$

where $\mathcal{L}_{\mathrm{fe}_i}$ is the feature embedding loss of the ith layer defined by (10) and β_i is the weight of the ith layer. The classification loss is defined as the summation of the cross-entropy loss over the source and target scenes, i.e., $\mathcal{L}_c = \mathcal{L}_{\mathrm{ce}}^{\mathcal{S}} + \mathcal{L}_{\mathrm{ce}}^{\mathcal{T}}$, where $\mathcal{L}_{\mathrm{ce}}^{\mathcal{S}}$ and $\mathcal{L}_{\mathrm{ce}}^{\mathcal{T}}$ are separately averaged over all samples within each scene. θ is the balancing parameter between the feature embedding loss and the classification loss. The detailed parameter settings will be discussed in Section IV.

In addition, because the architecture of the proposed CDVAE does not limit the dimension of the input sample, the samples of source and target scenes can be of the same dimension. Therefore, the algorithm can also be used for homogenous transfer learning.

C. Training and Testing Details

Two detailed points need to be clarified for training and testing.

1) In CDVAE, we use both source and target streams at the same time during training. The samples of source and target scenes are fed into the network together. However, in the test, we only use the target stream, and only the target scene samples are fed into the network.

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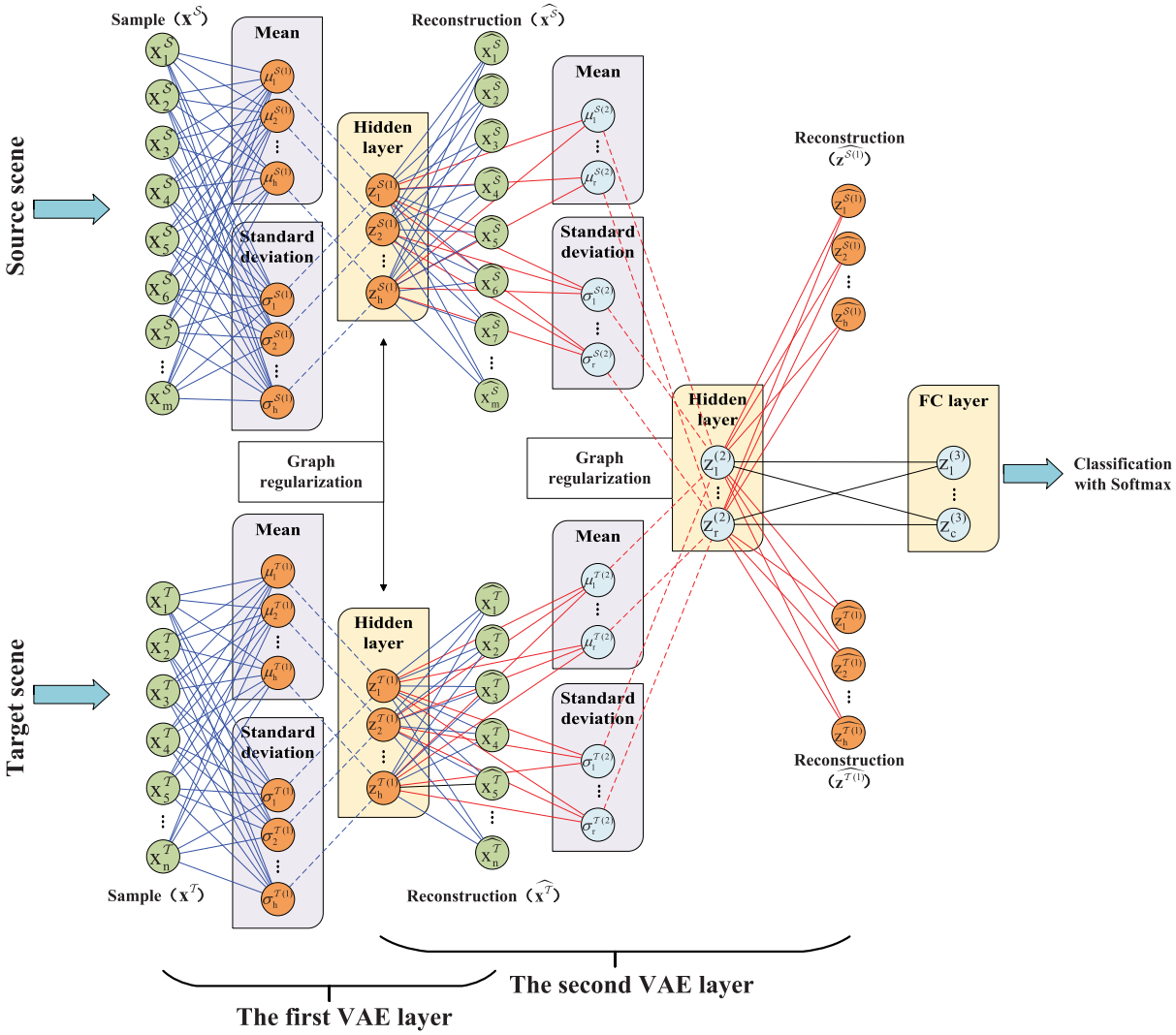


Fig. 4. Framework of the proposed CDVAE network.

2) Graph regularization loss is calculated over all training samples. In some extreme cases, the memory size cannot afford too many samples. Then, the following scheme can be adopted.
   a) In forward pass, divide source samples into batches and accumulate the total loss among all batches.
   b) Only perform one backward pass per epoch using the total loss.

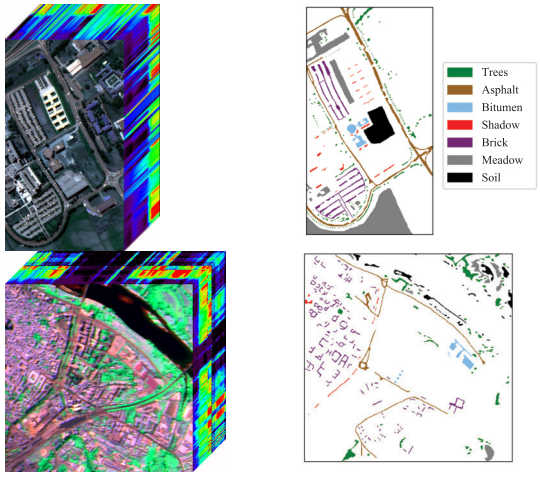
IV. Experiments

A. Datasets and Experimental Settings

In order to verify the effectiveness of the proposed algorithm, experiments are conducted on two real-world cross-scene HSI datasets.

1) *RPaviaU-DPaviaC Dataset:* It consists of two public HSI datasets: ROSIS Pavia University (RPaviaU) and DAIS Pavia Center (DPaviaC), where the former was captured by the ROSIS HSI sensor over the University of Pavia, Italy, while the latter was captured by DAIS sensor over the center of Pavia city, Italy. The source scene RPaviaU contains 103 spectral bands and 610 × 340 spatial pixels. The target scene DPaviaC contains 72 spectral bands and 400 × 400 spatial pixels. Both of them share seven land-cover classes. The land-cover classes and the corresponding number of samples are shown in Table I. The data cubes and ground-truth maps of the RPaviaU-DPaviaC dataset are shown in Fig. 5.
2) *EHangzhou-RPaviaHR Dataset:* It is composed of two public HSI scenes: EO-1 Hangzhou (EHangzhou) and ROSIS Pavia HR (RPaviaHR), where the former was taken with the EO-1 Hyperion hyperspectral sensor over Hangzhou city, Zhejiang, China, while the latter was captured by the ROSIS HSI sensor over Pavia city, Italy. The source scene EHangzhou dataset contains 198 spectral bands and 590 × 230 spatial pixels. The target scene RPaviaHR contains 102 spectral bands and 1400 × 512 spatial pixels. These two scenes share three land-cover classes. The number of samples in each class is listed in Table II. The details of land-cover classes can be seen in [21]. The data cubes and ground-truth maps of EHangzhou-RPaviaHR are shown in Fig. 6.

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(a) (b)

Fig. 5. Data cubes and ground-truth maps of RPaviaU-DPaviaC. (a) Data cubes. (b) Ground-truth maps. (Top) Source scene (RPaviaU). (Bottom) Target scene (DPaviaC).

TABLE I

NUMBER OF LABELED SAMPLES IN EACH LAND-COVER CLASS WITHIN THE RPAVIAU-DPAVIAC DATASET

Class		Labeled samples	
#	Name	Source scene	Target scene
1	Trees	3064	2424
2	Asphalt	6631	1704
3	Bitumen	1330	685
4	Shadow	947	241
5	Brick	3682	2237
6	Meadow	18649	1251
7	Soil	5029	1475

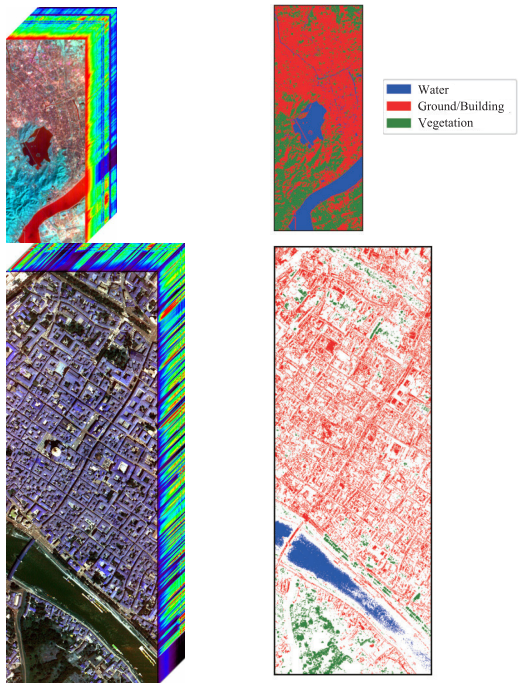
TABLE II

NUMBER OF LABELED SAMPLES IN EACH LAND-COVER CLASS WITHIN THE EHANGZHOU-PAVIAHR DATASET

Class		Labeled samples	
#	Name	Source scene	Target scene
1	Water	18403	22525
2	Ground/Building	77450	145416
3	Vegetation	40207	22961

In the cross-scene classification, the source scene contains abundant labeled training samples, while the target scene contains limited labeled training samples. In order to simulate the above conditions, we randomly select 400 labeled training samples from each class in the source scene, but only five labeled samples from each class in the target scene. The remaining target scene samples are included in the testing set. It should be noted that the samples of source scene are only used for transfer learning in the training procedure, i.e., the test process is only carried out in the target scene. After splitting training and testing sets, z-score normalization is introduced as a preprocessing.

The following six algorithms are described and compared, and their accuracies in the HSI classification are compared and analyzed.



(a) (b)

Fig. 6. Source and target cubes and ground-truth maps of EHangzhou-RPaviaHR. (a) Data cubes. (b) Ground-truth maps. (Top) Source scene (EHangzhou). (Bottom) Target scene (RPaviaHR).

1) *Spec:* It is short for classification with raw spectral feature. Here, support vector machine (SVM) with radial basis function (RBF) kernel is adopted as the classifier. This method is utilized as a baseline to see whether transfer learning delivers positive or negative transfer.
2) *Hybrid Spectral CNN (HybridSN):* It is an HSI classification algorithm proposed by Roy *et al.* [34], which is suitable for single-scene classification. In this algorithm, PCA is first used to reduce the spectral dimension, and then, it combines 3-D and 2-D CNNs to form a hybrid CNN architecture, thereby effectively extracting spectral and spatial information. When there are enough training samples, the algorithm shows very good performance in many HSI datasets.
3) *Target-Domain Variational Autoencoder (TDVAE):* It directly applies the original work of VAE to target-domain samples. With VAE, the data sampled from parameterized Gaussian distribution are used to reconstruct the input data. Compared with traditional feature extraction algorithms such as PCA and AE [30], VAE has better feature representation capabilities. For a fair comparison, two VAE layers are adopted for TDVAE, sharing the same settings with CDVAE.
4) *3-D-SRNet:* It is an HSI classification algorithm proposed by Jiang *et al.* [20], which is suitable for cross-scene classification. With the advantage of cross-scene transfer learning, 3-D-SRNet can reduce a large number of model parameters and achieve better classification performance. The algorithm first pretrains on source scene with enough training samples and then transfers

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TABLE III
COMPARED ALGORITHMS AND PARAMETER SETTINGS

Algorithm	Source training samples	Target training samples	Loss function (only for deep learning algorithms)	Main paramters
Spec	0	5 per class	-	• Penalty parameter of SVM: $C \in \{10^{-3}, 10^{-2}, \ldots, 10^{2}\}$; • Parameter of RBF kernel: $\gamma \in \{1, 2, 4, 8, 16, 32\}$;
HybridSN	0	5 per class	Cross-entropy Loss	• patch-size: 25×25; • Dimensions after PCA: 30; • Optimizer: Adam; • Learning rate: 0.001; • Training epochs: 100;
TDVAE	0	5 per class	VAE + Cross-entropy Loss	• The number of layers in VAE: 2; • Optimizer: Adam; • Learning rate: 0.001; • Training epochs: 1000; • The various parameters of the loss function: $\beta = 10$, $\theta = 1000$; • Output dimension: $r \in \{5, 10, 15, 20, 25\}$;
3-D-SRNet	400 per class	5 per class	Cross-entropy Loss	• patch-size: 5×5; • Optimizer: Adam; • Learning rate: 0.01; • Training epochs: 60;
DDNMF	400 per class	5 per class	-	• Graph regularization parameter: Dynamically update based on the number of iterations [21]; • Output dimension: $r \in \{5, 10, 15, 20, 25\}$;
CDVAE	400 per class	5 per class	Graph regularization + VAE + Cross-entropy Loss	• The number of layers in VAE: 2; • Optimizer: Adam; • Learning rate: 0.001; • Training epochs: 1000; • The various parameters of the loss function: $\beta = 10$, $\theta = 1000$, $\lambda = 0.001$; • Output dimension: $r \in \{5, 10, 15, 20, 25\}$;

the feature extraction network to the target scene for fine-tuning, so as to perform transfer learning.

5) *DDNMF:* It is a cross-scene HSI classification algorithm proposed by Chen *et al.* [35] in our previous work. The algorithm couples the training of source and target scenes through graph regularization so that the features of the same class in the two scenes remain similar. This algorithm shows excellent performance in feature alignment when there are sufficient labeled samples. However, being a shallow model, it may have lower classification performance when compared with deep learning models.
6) *CDVAE:* It is the algorithm proposed in this work. On the basis of VAE, sharing the classification layer and applying graph regularization result in knowledge transfer from source scene to target scene, so as to solve the problem of insufficient samples in the target scene. This algorithm is the focus of this work.

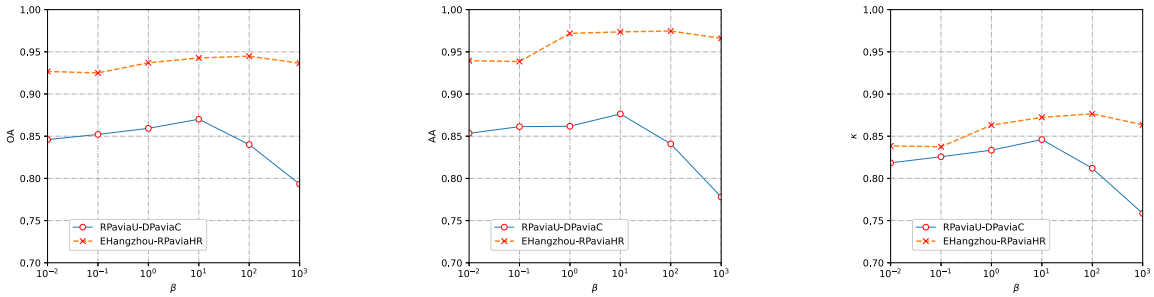
In the experiments of single-scene algorithms (Spec, HybridSN, and TDVAE), only target training samples are adopted for training. In the cross-scene experiments (3-D-SRNet, DDNMF, and CDVAE), both source and target training samples are fed into the model for training, and the feature dimensions of both source and target scenes are reduced to the same feature dimension r. The numbers of training samples, loss function, and parameter settings are listed in Table III. It is worth mentioning that our experiment assumes that source and target scenes are heterogeneous data, and the target scene has a small number of labeled samples. Among various transfer learning algorithms, some are only for homogenous transfer learning cases, and some others assume that the target scene does not contain any labeled sample. Thus, the settings of these algorithms [4], [5], [36]–[39] are not comparable with ours. Only 3-D-SRNet and DDNMF have the same experimental settings as the proposed one, and they are consequently included in the comparison.

B. Experiments on Parameter Settings of CDVAE

The loss function of CDVAE is obtained by summing up a number of different loss functions. The balancing parameters (β_i and θ in (11) and λ in (10)) are worthy of in-depth discussion. In this work, since CDVAE consists of two layers of two-stream VAE, we set $\beta_1 = 1$ and $\beta_2 = \beta$. In order to compare the impact of the choice of different parameters on performance, we adopt overall accuracy (OA), average accuracy (AA), and kappa coefficient (κ) to evaluate the performance of the algorithm. The following three balancing parameters are mainly analyzed.

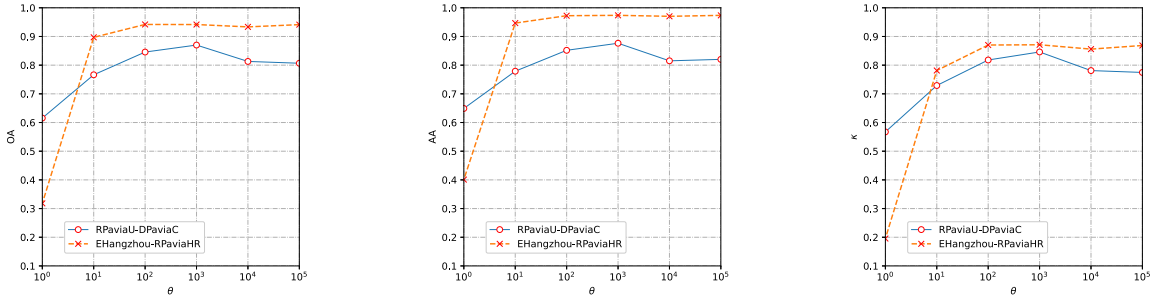
1) *Setting of the Balancing Parameter β Between $\mathcal{L}_{fe_1}$ and $\mathcal{L}_{fe_2}$ in CDVAE:* For the testing of the parameter β, we fix $r = 20$, $\lambda = 10^{-3}$, and $\theta = 10^{3}$. Experiments are performed on $\beta \in \{10^{-2}, 10^{-1}, \ldots, 10^{3}\}$. As shown in Fig. 7, the setting of β has little influence on the result, and when β takes a value near 10^{1}, the best result can be achieved in the two datasets. Thus, we select $\beta = 10$ in the subsequent experiments.
2) *Setting of the Balancing Parameter θ Between $\mathcal{L}_{fe}$ and $\mathcal{L}_{c}$:* We fix $r = 20$, $\lambda = 10^{-3}$, and $\beta = 10$. Experiments

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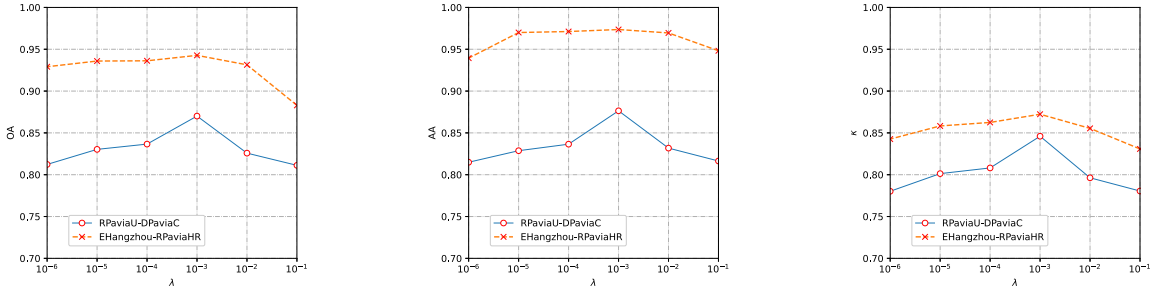
(a) (b) (c)

Fig. 7. Classification accuracy when taking different values of β. (a) OA. (b) AA. (c) κ.



(a) (b) (c)

Fig. 8. Classification accuracy when taking different values of θ. (a) OA. (b) AA. (c) κ.



(a) (b) (c)

Fig. 9. Classification accuracy when taking different values of λ. (a) OA. (b) AA. (c) κ.

are conducted with $\theta \in \{10^0, 10^1, \ldots, 10^5\}$. As shown in Fig. 8, when $\theta \in \{10^2, 10^3, 10^4\}$, the classification accuracy of the algorithm is not greatly changed. However, when $\theta < 10^2$, the classification accuracy of the algorithm begins to drop rapidly. The reason for this phenomenon is that the proportion of the classification loss is too small, which makes the algorithm unable to achieve a good balance between the transferability and the discriminability. When θ is near 10^3, the best result can be achieved. Thus, in the subsequent experiments, we set $\theta = 10^3$.

3) *Setting of the Balancing Parameter λ Between $\mathcal{L}_{vae}$ and the $\mathcal{L}_{gr}$ of Each Layer in CDVAE:* For testing the setting of λ, we fixed $r = 20$, $\theta = 1000$, and $\beta = 10$. Experiments are performed with $\lambda \in \{10^{-6}, 10^{-5}, \ldots, 10^{-1}\}$. As shown in Fig. 9, we find that too large or too small λ will lead to decrease on the accuracy, and $\lambda = 10^{-3}$ makes the best result. Thus, in this work, λ is set to 10^{-3}.

TABLE IV

OA, AA, AND κ OF EACH ALGORITHM IN THE RPAVIAU-DPAVIAC DATASET

Method	OA	AA	κ
Spec	0.8297	0.8106	0.7937
HybridSN	0.6503	0.6881	0.6076
TDVAE	0.7957	0.7904	0.7619
3-D-SRNet	0.7679	0.7844	0.7316
DDNMF	0.6240	0.6250	0.5431
CDVAE	**0.8701**	**0.8764**	**0.8460**

C. Comparison Among Different Algorithms

First, we conduct our study on the RPaviaU-DPaviaC dataset. The OA, AA, and κ are averaged from ten repeated experiments with random sampling, i.e., training and testing subsets are split ten times with random shuffling. For algorithms having feature dimension parameter r (DDNMF, TDVAE, and CDVAE), their accuracies with respect to r values are plotted in Fig. 10. It can be found that the best

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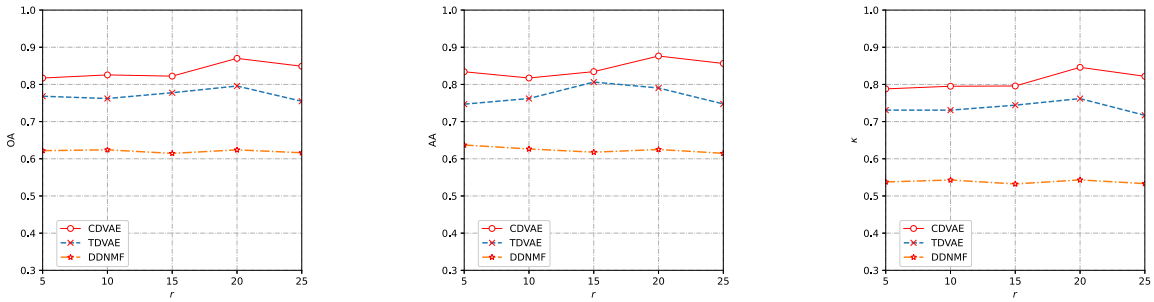


Fig. 10. Accuracies on the RPaviaU-DPaviaC dataset obtained by DDNMF, TDVAE, and CDVAE. (a) OA. (b) AA. (c) κ.

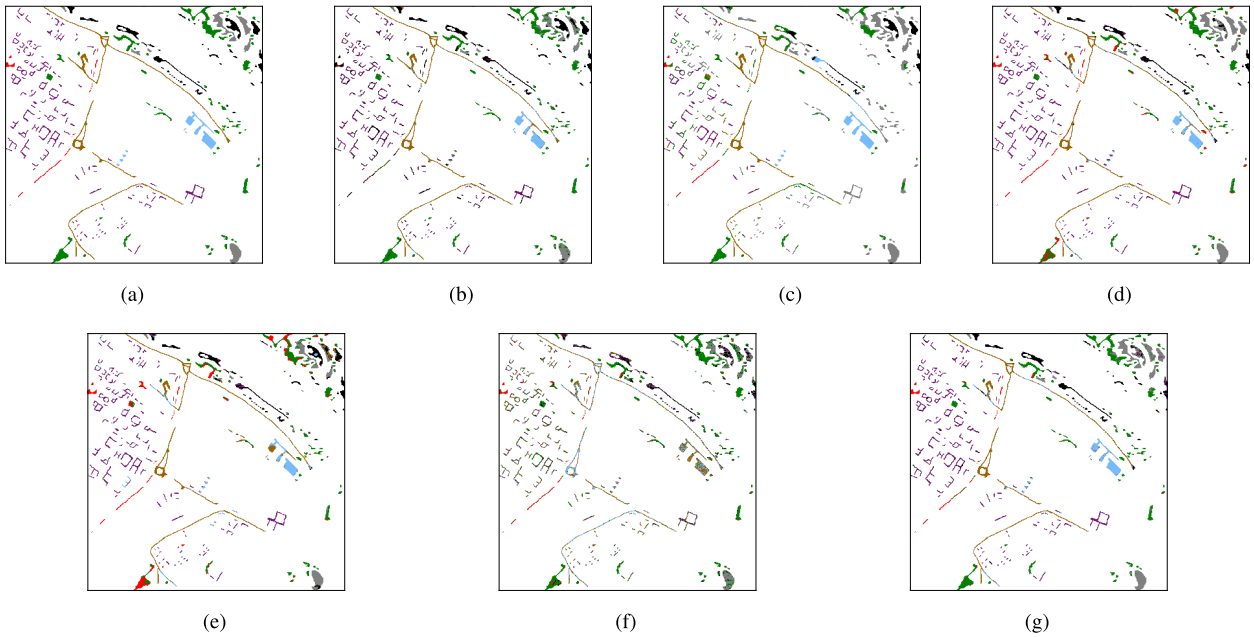


Fig. 11. Ground-truth and classification maps of RPaviaU-DPaviaC dataset. (a) Ground-truth map. (b) Classification map of Spec. (c) Classification map of HybridSN. (d) Classification map of TDVAE. (e) Classification map of 3-D-SRNet. (f) Classification map of DDNMF. (g) Classification map of CDVAE.

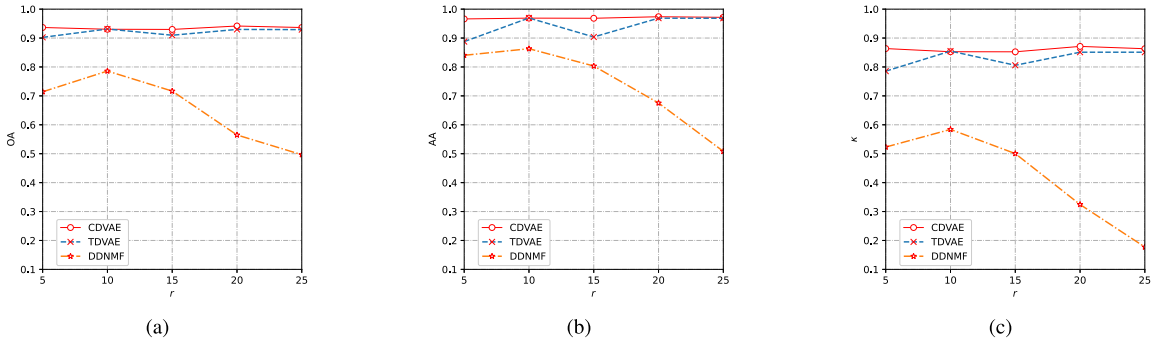


Fig. 12. Accuracies on the EHangzhou-RPaviaHR dataset obtained by DDNMF, TDVAE, and CDVAE. (a) OA. (b) AA. (c) κ.

accuracy of different algorithms may be achieved from different r values. OA, AA, and κ achieved from each algorithm are listed in Table IV. The ground-truth and classification maps of the RPaviaU-DPaviaC dataset are shown in Fig. 11.

The main experimental results are as follows.

1) As can be seen from Fig. 10 and Table IV, in the RPaviaU-DPaviaC dataset, the HybridSN algorithm produces poor performance when there are very few training samples. Since HybridSN has a complex architecture, it has a lot of parameters that need to be trained, overfitting may occur on this algorithm due to the lack of training samples. The DDNMF algorithm also cannot correctly classify the pixels of target scene, resulting in a very low classification accuracy. Similarly, the transfer

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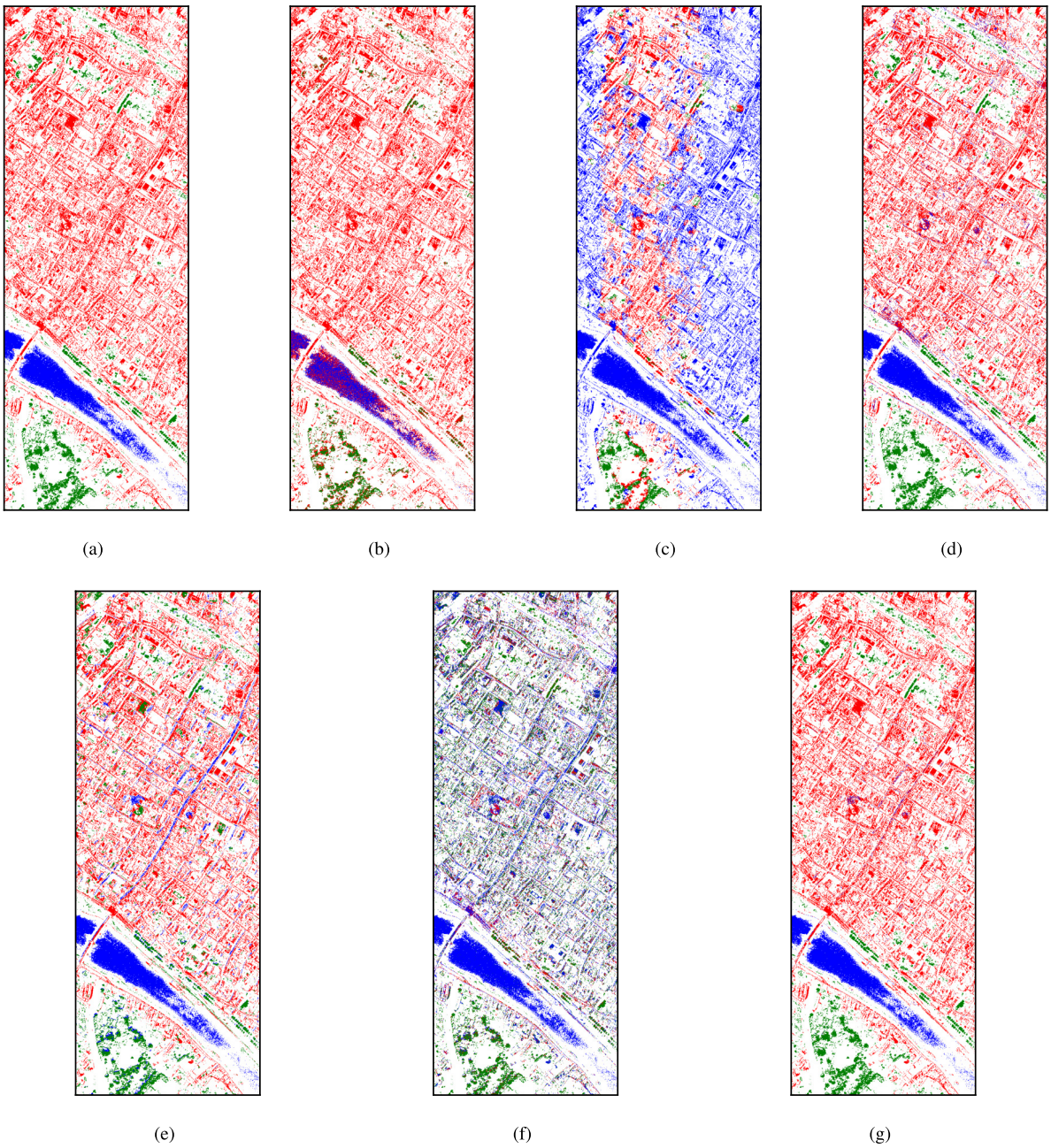


Fig. 13. Ground-truth and classification maps of EHangzhou-RPaviaHR dataset. (a) Ground-truth map. (b) Classification map of Spec. (c) Classification map of HybridSN. (d) Classification map of TDVAE. (e) Classification map of 3-D-SRNet. (f) Classification map of DDNMF. (g) Classification map of CDVAE.

learning network 3-D-SRNet achieves lower accuracy than the baseline. In other words, when there are not enough training samples in the target scene, all of the above algorithms deliver negative transfer. By contrast, CDVAE has obtained better results compared to TDVAE and Spec, which shows that CDVAE can effectively transfer the knowledge of source scene and improve the classification accuracy with very few training samples in the target scene.

2) As shown in Table IV, CDVAE achieves the highest accuracies among compared algorithms.

After the analysis and summary of the RPaviaU-DPaviaC dataset, we continue to carry out the experiments on the EHangzhou-RPaviaHR dataset. The accuracies of DDNMF, TDVAE, and CDVAE regarding r values are shown in Fig. 12. OA, AA, and κ of all compared algorithms are listed in Table V. The ground-truth and classification maps of the EHangzhou-RPaviaHR dataset are shown in Fig. 13.

As the data distribution of EHangzhou-RPaviaHR dataset is easier to be classified, the results are slightly different from those achieved from the RPaviaU-DPaviaC dataset. The specific results and analyses are as follows.

1) As shown in Fig. 12 and Table V, HybridSN, 3-D-SRNet, and DDNMF algorithms still perform poorly, which is similar to the case of RPaviaU-DPaviaC dataset.

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TABLE V

OA, AA, AND κ OF EACH ALGORITHM IN THE EHANGZHOU-RPAVIAHR DATASET

Method	OA	AA	κ
Spec	0.8554	0.7665	0.6577
HybridSN	0.6073	0.6816	0.4688
TDVAE	0.9312	0.9694	0.8549
3-D-SRNet	0.8390	0.8959	0.7019
DDNMF	0.7859	0.8631	0.5840
CDVAE	**0.9416**	**0.9737**	**0.8711**

TABLE VI

NUMBER OF TRAINABLE PARAMETERS IN CDVAE

Layer	Source stream	Target stream
First VAE layer	$151m + 100$	$151n + 100$
Second VAE layer	$152r + 50$	$152r + 50$
Fully connected layer (shared)	$rc + c$	
Total	$151m + 151n + 304r + rc + c + 300$	

2) As listed in Table V, the classification accuracy of TDVAE is much higher than that of the baseline method Spec, proving that VAE can achieve higher classification accuracy on this dataset.
3) It can be found in Fig. 12 and Table V that the performance of CDVAE is better than the remaining algorithms, which shows the effectiveness of combining VAE and transfer learning for HSI classification.

Finally, the number of trainable parameters in the proposed CDVAE network is analyzed in Table VI, where *m* and *n* are input dimensions of source and target scenes, respectively, *r* is the output dimension of second VAE layer, and *c* is the number of land-cover classes in the dataset.

V. CONCLUSION AND FUTURE WORK

In this work, we propose a heterogeneous transfer learning algorithm called CDVAE to solve a cross-scene HSI classification problem. Different from most existing heterogeneous transfer learning algorithms, the knowledge transfer is driven by the classification task through end-to-end training on a two-stream network. Specifically, the two-stream VAE maps samples of source and target scenes into a shared subspace, and knowledge transfer is delivered through subnet sharing and graph regularization. Through the cross-domain architecture and loss function, CDVAE guarantees both transferability and discriminability. In the experiments, CDVAE shows robustness to parameters, and more importantly, CDVAE achieves much better accuracies than the remaining algorithms.

It is worth mentioning that CDVAE only delivers spectral feature extraction. In our future work, hybrid spectral–spatial features are expected to be combined with CDVAE for a gain in classification performance.

REFERENCES

[1] S. Li, W. Song, L. Fang, Y. Chen, P. Ghamisi, and J. A. Benediktsson, "Deep learning for hyperspectral image classification: An overview," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 9, pp. 6690–6709, Sep. 2019.

[2] M. Ye, Y. Qian, J. Zhou, and Y. Y. Tang, "Dictionary learning-based feature-level domain adaptation for cross-scene hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 3, pp. 1544–1562, Mar. 2017.

[3] H. Zhang, Y. Li, Y. Jiang, P. Wang, Q. Shen, and C. Shen, "Hyperspectral classification based on lightweight 3-D-CNN with transfer learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 8, pp. 5813–5828, Aug. 2019.

[4] J. Lin, L. Zhao, S. Li, R. Ward, and Z. J. Wang, "Active-learning-incorporated deep transfer learning for hyperspectral image classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 11, no. 11, pp. 4048–4062, Nov. 2018.

[5] J. Geng, X. Ma, W. Jiang, X. Hu, D. Wang, and H. Wang, "Cross-scene hyperspectral image classification based on deep conditional distribution adaptation networks," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Jul. 2019, pp. 716–719.

[6] Y. Yao and G. Doretto, "Boosting for transfer learning with multiple sources," in *Proc. IEEE Comput. Soc. Conf. Comput. Vis. Pattern Recognit.*, Jun. 2010, pp. 1855–1862.

[7] Y. Xu *et al.*, "A unified framework for metric transfer learning," *IEEE Trans. Knowl. Data Eng.*, vol. 29, no. 6, pp. 1158–1171, Jun. 2017.

[8] M. Oquab, L. Bottou, I. Laptev, and J. Sivic, "Learning and transferring mid-level image representations using convolutional neural networks," in *Proc. IEEE Conf. Comp. Vis. Patt. Recogn.*, Oct. 2014, pp. 1717–1724.

[9] H. Chang *et al.*, "Unsupervised transfer learning via multi-scale convolutional sparse coding for biomedical applications," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 40, no. 5, pp. 1182–1194, May 2018.

[10] E. Tzeng, J. Hoffman, T. Darrell, and K. Saenko, "Simultaneous deep transfer across domains and tasks," in *Proc. IEEE Int. Conf. Comp. Vis.*, May 2015, pp. 4068–4076.

[11] E. Tzeng, J. Hoffman, K. Saenko, and T. Darrell, "Adversarial discriminative domain adaptation," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2017, pp. 7167–7176.

[12] J. Zhang, W. Li, and P. Ogunbona, "Joint geometrical and statistical alignment for visual domain adaptation," in *Proc. IEEE Conf. Comp. Vis. Pattern Recognit.*, 2017, pp. 5150–5158.

[13] M. Long, Y. Cao, Z. Cao, J. Wang, and M. I. Jordan, "Transferable representation learning with deep adaptation networks," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 41, no. 12, pp. 3071–3085, Dec. 2019.

[14] W. Kim and M. Crawford, "A novel adaptive classification method for hyperspectral data using manifold regularization kernel machines," in *Proc. Workshop Hypersp. Image Signal Process.*, Aug. 2009, pp. 1–4.

[15] S. J. Pan and Q. Yang, "A survey on transfer learning," *IEEE Trans. Knowl. Data Eng.*, vol. 22, no. 10, pp. 1345–1359, Oct. 2010.

[16] Z. Li, X. Tang, W. Li, C. Wang, C. Liu, and J. He, "A two-stage deep domain adaptation method for hyperspectral image classification," *Remote Sens.*, vol. 12, no. 7, pp. 1–18, 2020.

[17] Y. Zhang, W. Li, R. Tao, J. Peng, Q. Du, and Z. Cai, "Cross-scene hyperspectral image classification with discriminative cooperative alignment," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 11, pp. 9646–9660, Nov. 2021.

[18] X. Li, L. Zhang, B. Du, L. Zhang, and Q. Shi, "Iterative reweighting heterogeneous transfer learning framework for supervised remote sensing image classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 10, no. 5, pp. 2022–2035, May 2017.

[19] C. Zhong, J. Zhang, S. Wu, and Y. Zhang, "Cross-scene deep transfer learning with spectral feature adaptation for hyperspectral image classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 13, pp. 2861–2873, 2020.

[20] Y. Jiang, Y. Li, and H. Zhang, "Hyperspectral image classification based on 3-D separable ResNet and transfer learning," *IEEE Geosci. Remote Sens. Lett.*, vol. 16, no. 12, pp. 1949–1953, Dec. 2019.

[21] H. Chen, M. Ye, L. Lei, H. Lu, and Y. Qian, "Semisupervised dual-dictionary learning for heterogeneous transfer learning on cross-scene hyperspectral images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 13, pp. 3164–3178, 2020.

[22] G. E. Hinton and R. R. Salakhutdinov, "Reducing the dimensionality of data with neural networks," *Science*, vol. 313, no. 5786, pp. 504–507, 2006.

[23] D. P. Kingma and M. Welling, "Auto-encoding variational Bayes," 2013, *arXiv:1312.6114*.

[24] X. Hou, L. Shen, K. Sun, and G. Qiu, "Deep feature consistent variational autoencoder," in *Proc. IEEE Winter Conf. Appl. Comput. Vis. (WACV)*, Mar. 2017, pp. 1133–1141.

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[25] M. Belkin and P. Niyogi, "Laplacian eigenmaps and spectral techniques for embedding and clustering," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 14, 2002, pp. 585–591.
[26] M. Belkin and P. Niyogi, "Laplacian eigenmaps for dimensionality reduction and data representation," *Neural Comput.*, vol. 15, no. 6, pp. 1373–1396, 2003.
[27] D. Cai, X. He, J. Han, and T. S. Huang, "Graph regularized nonnegative matrix factorization for data representation," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 33, no. 8, pp. 1548–1560, Aug. 2011.
[28] L. Zhang, L. Zhang, D. Tao, X. Huang, and G. Xia, "Nonnegative discriminative manifold learning for hyperspectral data dimension reduction," in *Proc. Workshop Hypersp. Image Signal Process.*, 2013, pp. 1–4.
[29] D. Chen and P. Song, "Dual-graph regularized discriminative transfer sparse coding for facial expression recognition," *Digit. Signal Process.*, vol. 108, Oct. 2021, Art. no. 102906.
[30] C. Dong, T. Xue, and C. Wang, "The feature representation ability of variational autoencoder," in *Proc. IEEE 3rd Int. Conf. Data Sci. Cyberspace (DSC)*, Jun. 2018, pp. 680–684.
[31] M. Long, J. Wang, G. Ding, D. Shen, and Q. Yang, "Transfer learning with graph co-regularization," *IEEE Trans. Knowl. Data Eng.*, vol. 26, no. 7, pp. 1805–1818, Jul. 2014.
[32] M. Ye, W. Zheng, H. Lu, X. Zeng, and Y. Qian, "Cross-scene hyperspectral image classification based on DWT and manifold-constrained subspace learning," *Int. J. Wavelets Multiresolution Inf. Process.*, vol. 15, no. 6, pp. 1–16, 2017.
[33] C. M. Lee *et al.*, "Simultaneous non-negative matrix factorization for multiple large scale gene expression datasets in toxicology," *PLoS ONE*, vol. 7, no. 12, Dec. 2012, Art. no. e48238.
[34] S. K. Roy, G. Krishna, S. R. Dubey, and B. B. Chaudhuri, "HybridSN: Exploring 3-D–2-D CNN feature hierarchy for hyperspectral image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 17, no. 2, pp. 277–281, Feb. 2020.
[35] H. Chen, M. Ye, H. Lu, L. Lei, and Y. Qian, "Dual dictionary learning for mining a unified feature subspace between different hyperspectral image scenes," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Jul. 2019, pp. 1096–1099.
[36] J. Yang, Y.-Q. Zhao, and J. C.-W. Chan, "Learning and transferring deep joint spectral–spatial features for hyperspectral classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 8, pp. 4729–4742, Aug. 2017.
[37] J. Shen, X. Cao, Y. Li, and D. Xu, "Feature adaptation and augmentation for cross-scene hyperspectral image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 15, no. 4, pp. 622–626, Apr. 2018.
[38] J. Lin, R. Ward, and Z. J. Wang, "Deep transfer learning for hyperspectral image classification," in *Proc. IEEE Int. Works. Multimedia Signal Process.*, Oct. 2018, pp. 1–5.
[39] Z. Wang, B. Du, Q. Shi, and W. Tu, "Domain adaptation with discriminative distribution and manifold embedding for hyperspectral image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 16, no. 7, pp. 1155–1159, Jul. 2019.



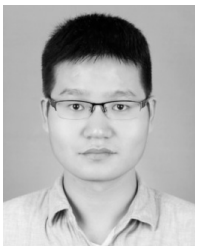
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